

Code Fast > Code Forward

HACKATHON SURVIVAL GUIDE

**Open Payments and all you need
to build the future of finance**

Why money should move the way information moves
on the web, and how developers build for it
with Open Payments and the Interledger Protocol.





You should be able to build for payments like you build for the web.

This guide is written for people who build. It explains why today's payment systems don't talk to each other, what the Interledger Protocol and Open Payments do about it, and how a payment actually flows through the Open Payments API, step by step.

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Short on time? Chapter 03 follows a single payment through the API in eight steps, from wallet address to confirmation. Start on page 21.



Open Payments Hackathon

BUILDING THE FUTURE OF FINANCE ➤

When

**16 and 17
October 2026**

Friday from 08:30
Saturday from 07:00

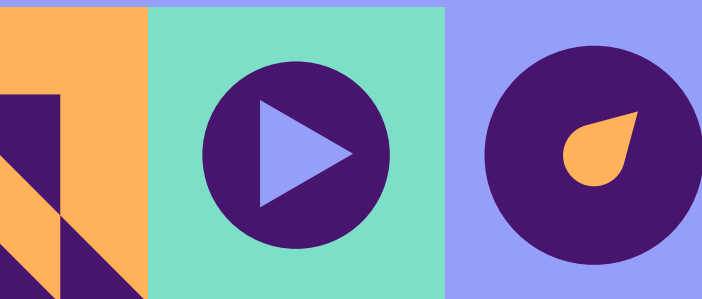
Where

**Complejo
Ruta N**

Calle 67 No 52-20, Piso 2, Torre A
Medellín, Colombia

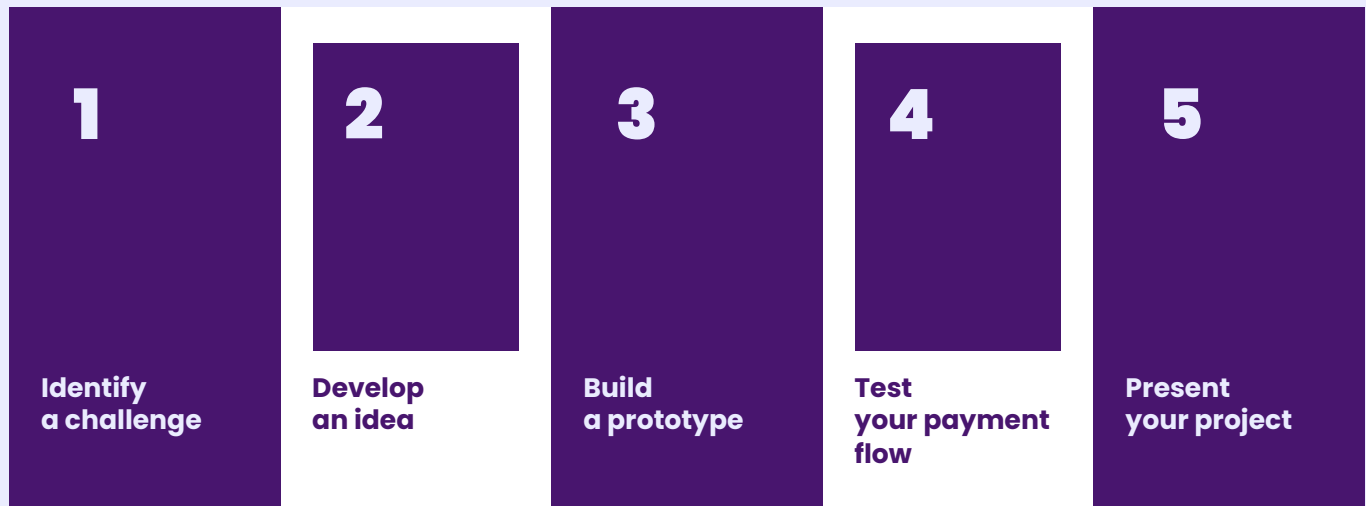
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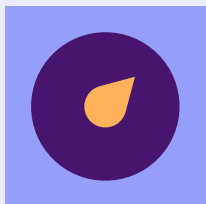


Hackathon goal

The goal at the Hackathon is to demonstrate how Open Payments can solve a real problem through a working prototype. During the hackathon, teams will:

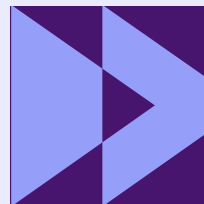


What solutions should demonstrate



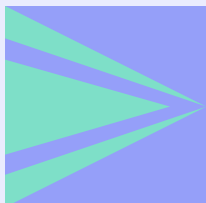
A clear problem

What challenge is being addressed.



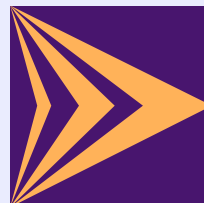
A user

Who benefits from the solution.



A working payment flow

Demonstration of payment in action.



A meaningful outcome

What changes because of the solution.

Tip for teams

A simple idea that works well is often stronger than a large project that tries to do too much.

Think about

Who is experiencing the problem?

What makes the current experience difficult?

What would a better experience look like?

How could Open Payments help?

Many of the best hackathon projects improve an existing experience rather than inventing something completely new.



Tools and resources

Everything you need is one click or scan away. Before building your application, try completing a payment flow using our guide: it will help you understand how the different parts of Open Payments work together.

Start here



**Payment flow
guide**



**Open
Payments
Documentation**

Build



**Interledger
Test Wallet**



**Example
repositories:
Open
Payments**

SDKs



JavaScript



PHP



Rust

Ask



**Community
Slack**



Take the free Open Payments course on Platzi

Take our beginner Open Payments course and start the hackathon already knowing the technology.

- Go to platzi.com and create a free account or log in.
- Access the course: Open Payments with Interledger.
- Complete the lessons and pass the final exam. If you fail the exam, you must wait 6 hours before retrying.
- Afterwards, join our community on Slack, introduce yourself in the #interledger-platzi channel and share your certificate (optional).

We also have a wealth of Open Payments resources, from documentation, to guides, to video series.



Course on Platzi

Mentors and timetable

Throughout the hackathon, a team of mentors will be on hand to help teams understand the Open Payments resources, sound board your ideas, and keep you motivated. They'll be available at a dedicated help desk and Slack channel, and will also be checking in on teams around the hack rooms.

Friday 16 October

08:30 – 10:00	Registration, breakfast
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10:00 – 10:30	Welcoming ceremony
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10:30 – 14:00	Session 1: Hacking
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14:00 – 15:00	Lunch
----------------------	-------

15:00 – 17:30	Session 2: Hacking
----------------------	--------------------

17:30 – 18:00	Departures
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Saturday 17 October

07:00 – 08:00	Registration, breakfast
----------------------	-------------------------

08:00 – 11:00	Session 3: Hacking
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11:00 – 12:00	Final code submitted to hack room folder/repositories; code review; teams practice presentations
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12:00 – 13:00	Round 1 Judging
----------------------	-----------------

13:00 – 14:30	Lunch
----------------------	-------

14:30 – 15:30	Finalists announced; Round 2 Judging
----------------------	--------------------------------------

15:30 – 16:30	Networking and refreshments
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16:30 – 17:30	Winners' announcement and closing ceremony
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17:30 – 18:00	Departures
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Overnight

Sleep is for Sunday.

Ruta N closes at 18:00, but the clock doesn't stop. All teams are encouraged to keep building from home overnight, and a virtual mentor team will be available for any questions.

Awards

Every solution will be considered for the Financial Inclusion, Innovation, and Impact Awards as part of the judging process. After the final presentations, all attendees will be asked to vote for their favorite solution for the Community Award via interactive voting.

Puntored Award: Financial Inclusion Award

The project that best advances financial inclusion by making digital payments more open, accessible, and equitable.

FIND Innovation Award

The project that demonstrates the most original thinking and innovative use of Open Payments, showing creativity to build the future of finance.

Ruta N Impact Award

The project with the strongest potential for real-world impact using Open Payments.

This Week in Fintech Community Award

The team or project that best embodies the Interledger community spirit. Voted for by all participants, recognizing the power of community in building networks for fintech.

How judging works

Round 1

Every team presents to a judging panel. This takes place in the hack rooms on Saturday at 12:00, after all participants have submitted their final presentations and code. There will be 4 rooms, each with a minimum of 2 judges. Teams present to the judges in their assigned room. Once all teams in a room have presented, the judges deliberate and select the teams to progress to Round 2, based on each team's cumulative score against the judging criteria.

Round 2

The 12 teams progressing to Round 2 are announced in the auditorium and immediately present their solutions to a new judging panel, which includes representatives from each award category. All participants also vote for their favorite solution, for the This Week in Fintech Community Award. Once all Round 2 presentations are complete, the judging panel convenes to select the overall award winners. All participants are then invited back into the auditorium for the winners' announcement.

Judging criteria

Criteria	What we're looking for
Problem definition	Is the challenge clear and meaningful?
Use of Open Payments	Does the project demonstrate a working payment flow?
User experience	Does the solution create value for users?
Technical quality	Does the prototype work reliably?
Presentation	Can the team clearly explain what they built and why it matters?
Innovation	How creative and original is the solution's approach to payments?
Inclusion	How does this solution make payments easier, more accessible or inclusive for underserved users?
Impact	What difference could this solution make for users or communities, and how clearly is that explained?

Preparing your submission

Submissions include

- › Presentation slides
- › Link to demo
- › Link to public GitHub repo

Optional materials

- › Architecture diagrams
- › Mockups
- › Technical documentation
- › Videos

Your presentation should explain

- › The challenge
- › The user
- › The solution
- › How Open Payments was used
- › Key learnings

Your demo should show

A working demonstration and recording of the payment flow, clearly showing:

- › Payment request
- › Authorization
- › Payment execution
- › Successful completion

FAQs

Do I need prior experience with Open Payments?

No. The hackathon is designed for both beginners and experienced developers.

Do I need to build a complete product?

No. Focus on demonstrating a clear payment experience and a working prototype.

What should I build first?

Get one successful payment working. Everything else can be built from there.

Can I use AI?

Yes, following our AI policy for hackathons.

Where can I get help?

Use the documentation, community channels, workshops, mentors, and other participants throughout the hackathon.

Do I need to use Open Payments to qualify for prizes?

Yes. Open Payments integration is a mandatory requirement for all hackathon submissions. Projects that don't incorporate the Open Payments API won't be eligible for judging or prizes.

Does my GitHub repository need to be public?

Yes. All project repositories must be publicly accessible so judges and code reviewers can access, test, and evaluate your code. Private repositories won't be considered for judging unless otherwise approved by the hackathon organizers.

Code of conduct

The Interledger Foundation is dedicated to providing a harassment-free event experience for everyone, regardless of gender, gender identity and expression, sexual orientation, disability, physical appearance, body size, race, age, religion or lack thereof. We do not tolerate harassment of event participants in any form. Sexual language and imagery are not appropriate for any event venue, including talks. Event participants violating these rules may be sanctioned or expelled from the event at the discretion of the ILF staff.



Full FAQ



Code of
Conduct



AI policy

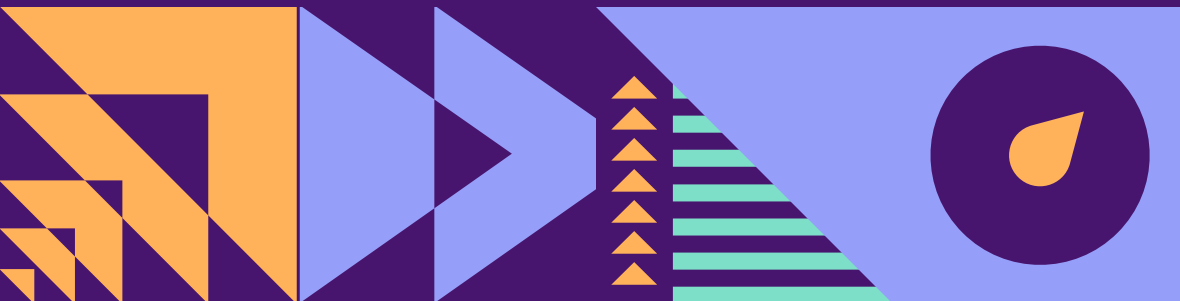
01

A FRAGMENTED FINANCIAL SYSTEM >

Thousands of banks, wallets and payment processors, and almost none of them talk to each other. Why moving money is still harder than moving information.

In this chapter >

Systems that rarely interoperate	12
The need for open standards	13
What Open Payments and ILP aim to change	14



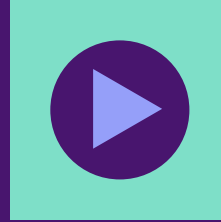
Systems that rarely interoperate

Today's global financial ecosystem is made up of thousands of independent systems: banks, digital wallets, mobile money services and payment processors, all operating under different technical and legal rules.

Even within a single country, moving money between services, for example from a neobank to a ride-sharing app wallet, can be impossible unless both companies have negotiated a custom bridge.

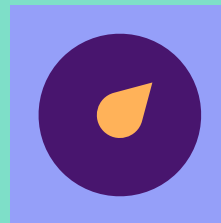
That makes innovation slow, expensive, and accessible only to the largest players.

Each system works on its own. Between them: a custom bridge, a private partnership, or nothing at all.



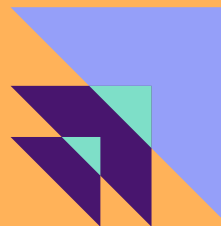
Across borders

A wallet in one country can't easily send funds to a bank account in another, even if both are online.



Across providers

Sending money between different providers often requires custom integrations, private partnerships, or even manual processes.



For developers

Every payment method must be integrated separately, accounting for the unique aspects of each provider.

The need for open standards

This lack of interoperability leads to serious challenges for individuals, businesses and entire economies. We have built a digital economy on top of a financial system that is still fragmented by geography, business models, and outdated infrastructure.

Cross-border payments are slow and expensive

They often take days and cost users high transaction and conversion fees. This especially hurts migrant workers sending money home.

Micropayments are not economically viable

Card-based infrastructure carries minimum fees and batch processing limitations that make very small payments impractical.

Startups and developers are locked out

Innovation requires the budget to integrate with dozens of separate APIs and to navigate complex compliance.

Digital financial inclusion suffers

Particularly in regions where local financial systems are isolated from global networks.

If you're a developer, this is a real problem

You shouldn't need to partner with a global payment provider just to send \$5 to a friend.

You may have an amazing idea for a payment-enabled application, but executing it means facing a minefield of closed APIs, legal restrictions, and legacy complexity. You shouldn't have to build a separate integration for every country, wallet, or bank.

What Open Payments and ILP aim to change

The Interledger Protocol (ILP) allows value, such as money, to move across different financial networks, just as the Internet allows information to move across devices and platforms. It ensures that payments can be routed securely from one system to another, even if they use different underlying technologies or currencies.

Open Payments builds on ILP by providing a standardized set of web APIs that developers use to create, authorize, and send payments through these connected systems.

Together they let money move between systems the way email moves between providers or websites work across browsers. Just as HTTP, TCP/IP and DNS made the Internet accessible to all, Open Payments and ILP lay a new foundation for an Internet of Opportunity, where anyone, anywhere, can build or access financial tools that connect to a global payment network.

Payments can become

Interoperable across systems and borders

Programmable using modern APIs

User-consented and secure by default

**Inclusive for people
and providers everywhere**

**You're not just building a payment app.
You're helping build an Internet of Opportunity.**

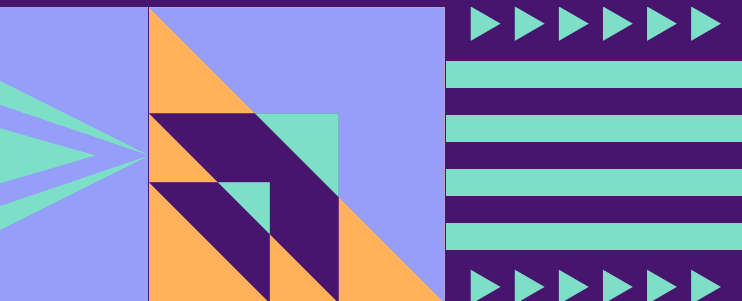
02

THE INTERLEDGER TECHNOLOGY STACK ➤

Like the Internet, Interledger is built in layers. Each one does a specific job, and each can evolve on its own. Here is what sits where.

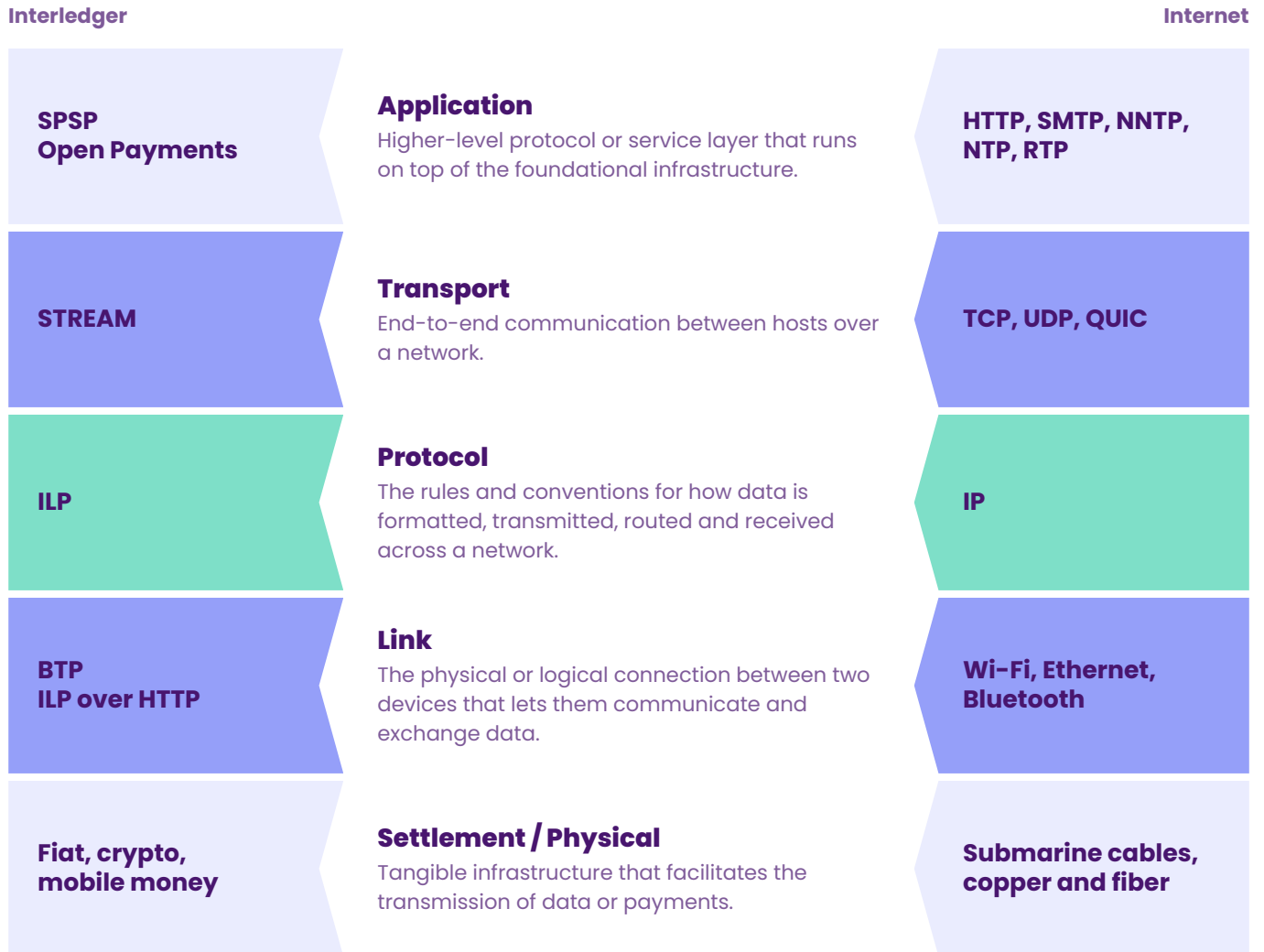
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Five layers, side by side with the Internet

Just like the Internet has layers (HTTP, TCP, IP), ILP has its own stack. The whole stack sits on top of a settlement layer: the existing systems that Interledger connects. That layer isn't strictly part of ILP, but it is an indispensable part of the model.



Where you build >

The application layer. Open Payments is the API you call: wallet addresses, grants, incoming payments, quotes and outgoing payments.

What you never touch

STREAM, ILPv4, the link layer and settlement. You won't write packet logic or operate connectors; ILP handles that when your outgoing payment is created.

What each layer does

Application layer

Where developers interact directly.

SPSP (Simple Payment Setup Protocol) facilitates the exchange of ILP connection credentials needed to set up a payment.

Open Payments is a full-featured API layer for account providers to expose secure, standardized payment functionality.

Transport layer: STREAM

Segments and manages payments and metadata over a secure connection. It enables streaming money in real time and multiplexing several streams at once.

Protocol layer: ILPv4

The core of the stack. It describes the expected structure and behavior of ILP packets, and how they should be handled.

Link layer

Defines how two systems communicate securely. Examples include ILP over HTTP and BTP (Bilateral Transfer Protocol).



The ILP vision

Enable value to move as easily, efficiently and universally as data does on the Internet. Where TCP/IP lets packets of data travel across disconnected computer networks, ILP lets packets of money travel across disconnected financial systems.

It is a bridge layer that abstracts the differences between systems and makes sure the payment is routed, delivered and confirmed securely.

How ILP actually works

At its core, ILP is a messaging protocol for payments. It allows money to be sent from one party to another even if they use different financial providers, networks, or currencies.

To do this, ILP breaks payments into small packets, which are routed through **connectors**: intermediary nodes that pass value along a chain until it reaches the recipient.

Universal execution

Conditional transfers secure payments across multiple hops. Each connector only needs to trust its direct peers.

Currency-agnostic

ILP can be used with local currencies, cryptocurrencies, and any payment method that a connector node is willing to bridge.

What ILP does when a payment is initiated

- › Splits the payment into small, independently routed packets.
- › Passes each packet through a series of connectors, the trusted intermediaries.
- › Handles currency conversions, network compatibility and settlement conditions.

This lets a user on one system, like a mobile wallet, pay someone on a completely different system, like a bank account or a blockchain wallet, without those systems ever communicating directly with each other.



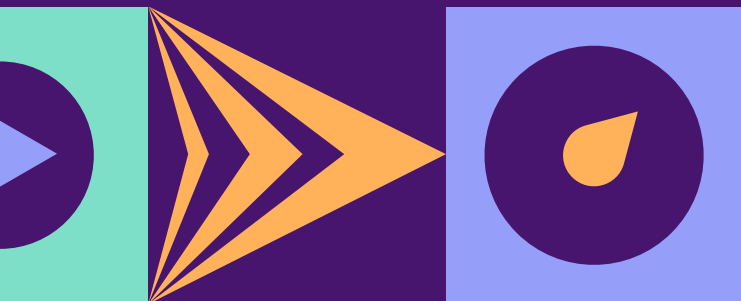
03

BUILDING WITH OPEN PAYMENTS➤

One API flow instead of a custom integration per provider.
What the API lets you do, and a payment followed step by step
from wallet address to confirmation.

In this chapter >

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Sequential, but independent	23



One unified API flow

Open Payments sits at the top of the Interledger stack, in the application layer. It builds on ILP and STREAM by providing a RESTful API interface: a set of secure, standardized APIs that allow account providers such as wallets, banks or mobile money platforms to expose payment functionality in a programmable and interoperable way.

Developers and apps can

- › Discover account information
- › Create and manage payment requests
- › Request real-time quotes
- › Obtain user authorization (via GNAP)
- › Initiate and track outgoing payments

Instead of one integration per provider

Developers use Open Payments to initiate and complete payments through one API flow. Complex financial operations become a predictable, modular process that supports real-time, user-consented transactions.

Every step below maps to its own API resource, so each can be created, reviewed and approved on its own.

Your platform

Sender and receiver

Requests payment from a user



The receiver defines the amount they want to receive

Presents a precise, costed quote



The sender sees exactly how much they'll be charged

Obtains permission to proceed



The user authorizes the action

Executes a real-time transfer over Interledger



The payment executes with complete transparency

Peer-to-peer flow, step by step

In this example the sender initiates the payment.

The steps are modular, and each one maps to a separate API resource.

1

Discover wallet capabilities

Every participating account provider hosts wallet addresses for its users in the form of a URL, for example

`https://wallet.example.com/alice`

When queried, the endpoint returns:

- › Supported currencies
- › Public account metadata
- › API endpoints for next steps: grants, incoming payment, quote, outgoing payment

This is the entry point for initiating a payment.

2

Request an incoming payment grant

The sender's app requests an incoming payment grant from the **recipient's** wallet authorization server. The request specifies:

- › The type of resource: incoming-payment
- › The actions the app wants to perform, e.g. create and read

Open Payments uses GNAP to negotiate and authorize every grant. For incoming payment grants this happens behind the scenes. The app receives an access token.

3

Create an incoming payment

The sender's app requests that an incoming payment resource be created on the recipient's wallet account. The request includes:

- › The amount to be received
- › An optional expiration date or description
- › Reference metadata, like an order ID or invoice number

This payment request can be shared with the sender.

4

Request a quote grant

The sender's app requests a quote grant from the **sender's** wallet authorization server. The request specifies:

- › The type of resource: quote
- › The actions the app wants to perform, e.g. create and read

Again via GNAP, and again behind the scenes. The app receives an access token in return.

Peer-to-peer flow, continued

Quotes let users approve payments with full visibility of costs and conversions before anything is sent. No payment proceeds without the sender's explicit consent.

5

Generate a quote

The sender's app requests a quote resource on the sender's wallet account, with identifying information for the sender and the URL of the incoming payment resource.

The sender's wallet uses the incoming payment details to create a quote that determines:

- › How much the sender must pay in their own currency
- › Estimated exchange or network fees
- › The delivery amount in the receiver's currency

6

Request an interactive outgoing payment grant

GNAP supports obtaining consent through interactive grants. The sender's app requests one from the sender's wallet authorization server, including:

- › The type of resource: outgoing-payment
- › The actions the app wants to perform, e.g. create and read
- › Information on how to begin the interaction

This starts a secure, out-of-band flow where the sender approves the requested action, e.g. "send \$10 to merchant X". The app receives an access token.

7

Initiate the outgoing payment

The sender's app requests that an outgoing payment resource be created on the sender's wallet account. This final resource:

- › References the quote or incoming payment resources
- › Debits the sender's account
- › Transfers funds using the Interledger network

This step completes the payment securely, with confirmation and traceability.

8

Monitor and confirm the transaction

Both sender and recipient can monitor payment status through the Open Payments API. Each resource includes metadata, status updates and timestamps to track:

- › Whether the payment has been received
- › Whether the full amount was delivered
- › When it was completed

This visibility is especially useful for automation, receipts, or billing.

Sequential, but independent

These components are used in order, but none of them depends on the others being triggered.

How the components work together

- ▶ You can create and share an incoming payment before the sender even knows about it.
- ▶ A quote can be created without triggering a payment.
- ▶ Grants are scoped to specific actions, so one token doesn't authorize everything.

Why resource-based design matters

In traditional systems, payment APIs often bundle authorization, pricing and execution into a single request, which makes it hard to understand, test, or modify individual parts.

Open Payments takes the opposite approach: resource separation lets each step in the payment be created, reviewed, approved and executed independently.

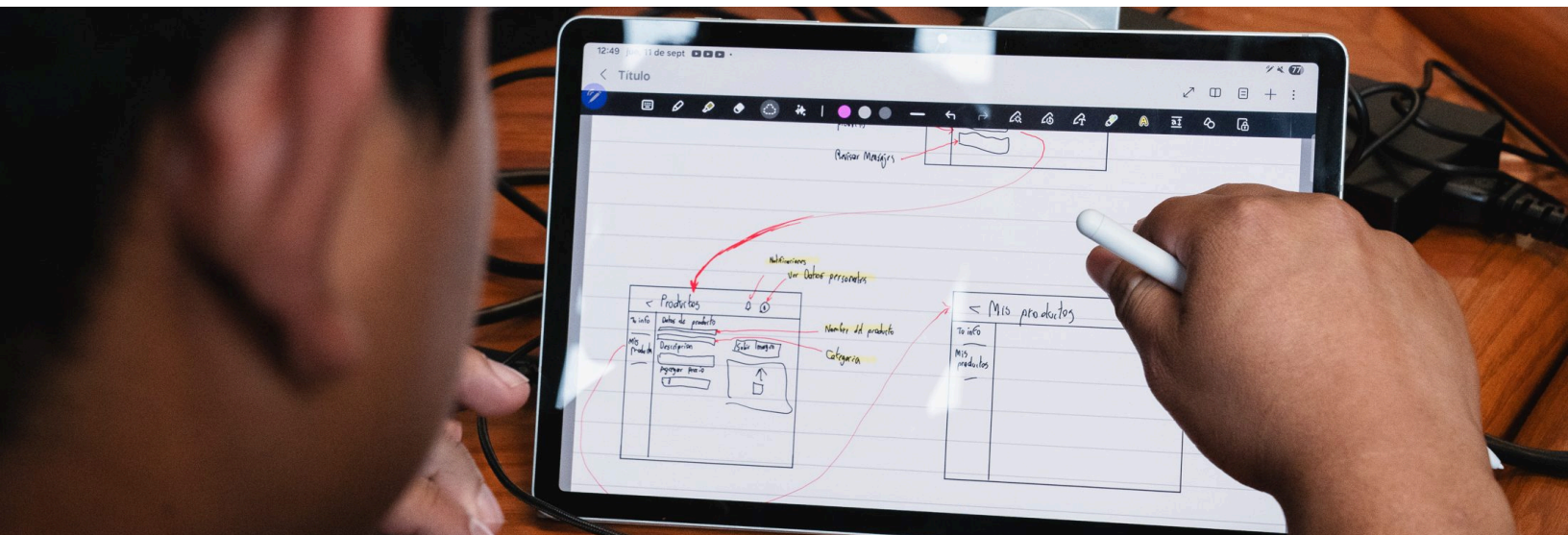
That makes Open Payments a natural fit for applications that require explicit consent, pre-transaction quoting, fine-grained control, and secure audit trails.

Simplifies integration and debugging

Makes flows more transparent for users

Lets apps mix and match components for different use cases

If one step fails, you know exactly where and why.



04

ILP UNDER THE HOOD»

Open Payments defines the what.
ILP defines the how.
Once a user has approved a payment,
this is how the money actually travels.

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Core ILP features that enable interoperability

Open Payments gives developers a clean, modular way to initiate payments, authorize actions and track transactions through APIs. But once a user has approved a payment and the transaction is ready to be sent, how does the money actually get from sender to receiver, especially across different systems or currencies? That is ILP's job.

Packetized payments

Like the Internet breaks data into packets, ILP breaks payments into units that can be routed independently. This lets both microtransactions and high-volume transfers move efficiently.

Connector-based routing

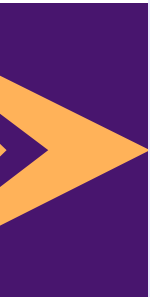
Connectors relay payment packets between senders and receivers, and can convert between currencies or ledgers. Each connector only needs to trust its neighbors, not the entire network.

Conditional transfers and fulfillment

Transfers are prepared conditionally and only finalized when the receiver confirms receipt. If confirmation fails, no funds move. Both parties are protected from partial or failed payments.

Multi-currency, multi-hop

ILP is currency-agnostic. Sender and receiver can use different currencies, with connectors performing exchanges along the way. That is what makes true cross-border, cross-network payments possible.



Open Payments defines the **what: who sends, how much, to whom. ILP defines the **how**: value moving securely across systems that don't natively interoperate.**

What this makes possible

With ILP operating under the hood, Open Payments can support use cases that would otherwise require massive custom infrastructure.

Streaming payments

For digital media or live events, paid continuously as they happen.

Cross-border donations

From users in one country to creators in another.

Split payments

One transaction routed to multiple recipients.

Micropayments

Amounts too small for traditional card networks to handle cost-effectively.

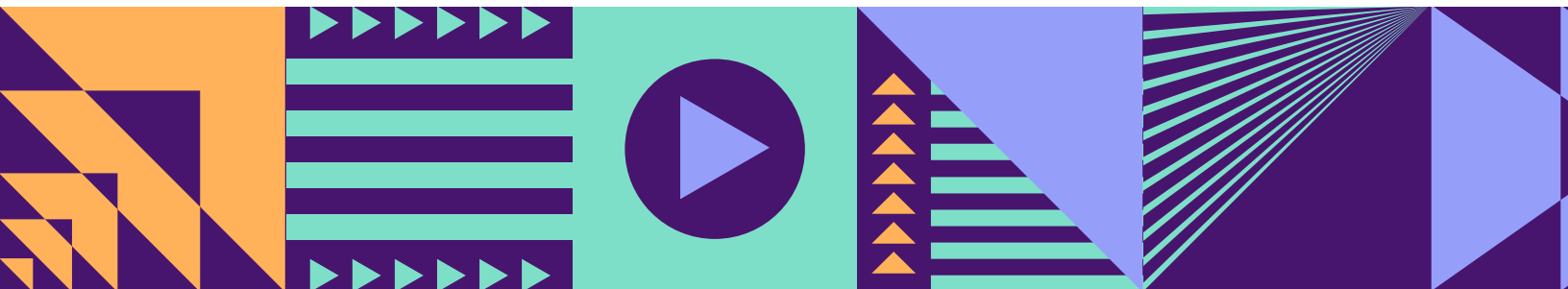
Your app doesn't need to implement any of this complexity. You create an outgoing payment through the Open Payments API, and ILP makes sure it gets where it needs to go.

Why this matters in practice

As a developer, you don't interact with ILP directly. You won't write packet logic or operate connectors. But understanding what ILP is doing in the background helps you:

- › Design apps that work across wallets, currencies and countries
- › Articulate how your solution is interoperable, not vendor-locked
- › Build payment features that scale beyond a single provider

ILP is the invisible layer that lets your Open Payments-powered app work globally, safely and efficiently, without locking you into any single provider, ledger, or currency.



Terms you'll meet while building

Wallet address

A URL hosted by an account provider that identifies a user's account and exposes the endpoints for grants, incoming payments, quotes and outgoing payments.

Account servicing entity

The wallet, bank or mobile money provider that holds the account and runs its authorization and resource servers.

Grant

Permission for an app to act on an account, scoped to a resource type and a set of actions. Returned as an access token.

GNAP

Grant Negotiation and Authorization Protocol. The protocol Open Payments uses to negotiate grants, including interactive grants that require the user's consent.

Incoming payment

A resource on the receiver's account that describes what they want to receive: amount, optional expiry or description, and reference metadata.

Quote

A resource on the sender's account that fixes how much the sender pays, the estimated fees, and how much the receiver will get, before anything is sent.

Outgoing payment

The resource that debits the sender's account and moves funds over Interledger. Creating it requires an interactive grant.

Connector

An intermediary node in the Interledger network that relays payment packets and can convert between currencies or ledgers. It only needs to trust its direct peers.

ILPv4

The current version of the Interledger Protocol. It defines the structure and behavior of ILP packets.

STREAM

The transport protocol that segments payments and metadata over a secure connection and enables real-time streaming of money.

SPSP

Simple Payment Setup Protocol. Exchanges the ILP connection credentials needed to set up a payment.

BTP / ILP over HTTP

Link-layer protocols that define how two Interledger systems communicate securely with each other.

Keep going >

- Open Payments documentation: openpayments.dev
- Interledger Foundation: interledger.org
- Hear the founders' story: Stefan Thomas and Evan Schwartz on the FIM Podcast, "Reimagine global payments with the creators of Interledger" (English).
- Watch Stefan Thomas, ILP co-creator: "How I learned to Love Packetisation".



CODE FAST ➤ CODE FORWARD

Open Payments Hackathon
Ruta N, Medellín · 16 and 17 October 2026

Open Payments
HACKATHON
BY Interledger

